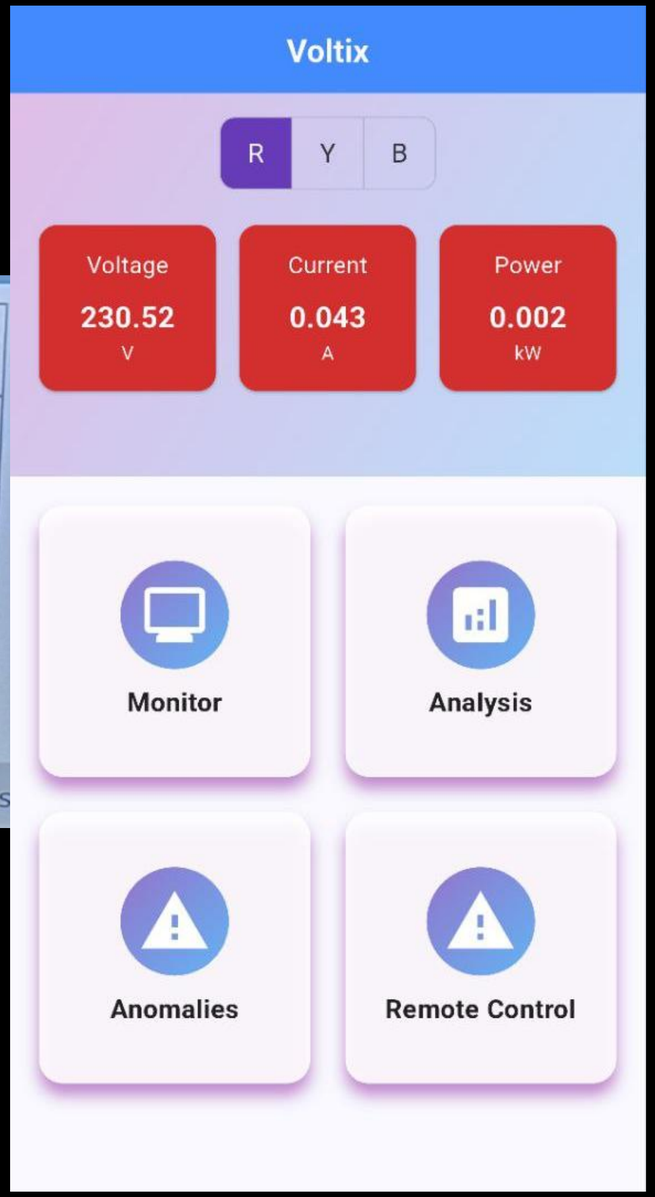
Voltix Meter

Team Members

- Saurabh Kumar (22B3905)
- Aryan Tamboli (22B3927)
- Sarthak Niranjan (22B3929)
- Cherish Jain (22B3937)
- Ayush Gaur (22B3973)

Phase	Vrms(V)	Irms(A)	P(W)	Q(VAR)	S(VA)	PF
R	239.29	0.51	12.11	-118.05	123.02	0.10
Y	240.80	0.38	53.22	58.34	92.51	0.58
B	238.39	0.00	0.02	0.01	0.00	0.00
Neutral	0.00	0.07	-	-	-	-

Energy | Power | Failure Log | Get QR | History | V - I | Home | Harmonics



An **IoT-based 3-phase smart energy meter** for **Industries** to remotely monitor, control supply, and analyze power quality, **solving** issues of manual monitoring and inefficient energy usage.

Status of Project

Key specifications and requirements proposed at beginning of semester	Status at end of semester	Remarks/Justification
Real-time monitoring of voltage, current, power, and harmonics	Complete	Working Correctly
30+ days cloud data storage with mobile app access	Complete	Working Correctly
Reliable isolation and protection for high-voltage systems	Complete	Working Correctly
Remote control of 3 phases of supply using relay	Complete	All 3 phases will be activated or deactivated simultaneously using our **Android app - Voltix**
Improved software stack	Complete	The screen display and app work seamlessly without lag
Power Backup in case Phase connected to RPI goes out	Not working	Battery Backup built and tested but battery is not getting charged so Backup is not working

Photographs of Final Prototype

Phase	Voltage THD (%)	Current THD (%)	Frequency (Hz)
R	1.77	43.88	49.91
Y	1.70	59.97	49.90
B	1.69	0.00	49.90

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Harmonics Page

Home Page

Phase	Vrms(V)	Irms(A)	P(W)	Q(VAR)	S(VA)	PF
R	239.29	0.51	12.11	-118.05	123.02	0.10
Y	240.80	0.38	53.22	58.34	92.51	0.58
B	238.39	0.00	0.02	0.01	0.00	0.00
Neutral	0.00	0.07	-	-	-	-

Month: April Year: 2025 Phase R Phase Y Phase B

Load Data

Energy

Energy (kWh)

0.06 0.05 0.04 0.03 0.02 0.01 0.00

1-Apr 05-Apr 10-Apr 15-Apr 20-Apr 25-Apr 30-Apr

gy | Power | Failure Log | Get QR | History | V - I | Home | Harmonics

History Page

Energy Page

Phase R Phase Y Phase B

Phase R	Phase Y	Phase B
0.06113	0.01277	0.00001

Energy (kWh)

0.06 0.04 0.02 0.00

00:00 03:00 06:00 09:00 12:00 15:00 18:00

Energy | Power | Failure Log | Get QR | History | V - I | Home | Harmonics

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V-I Page

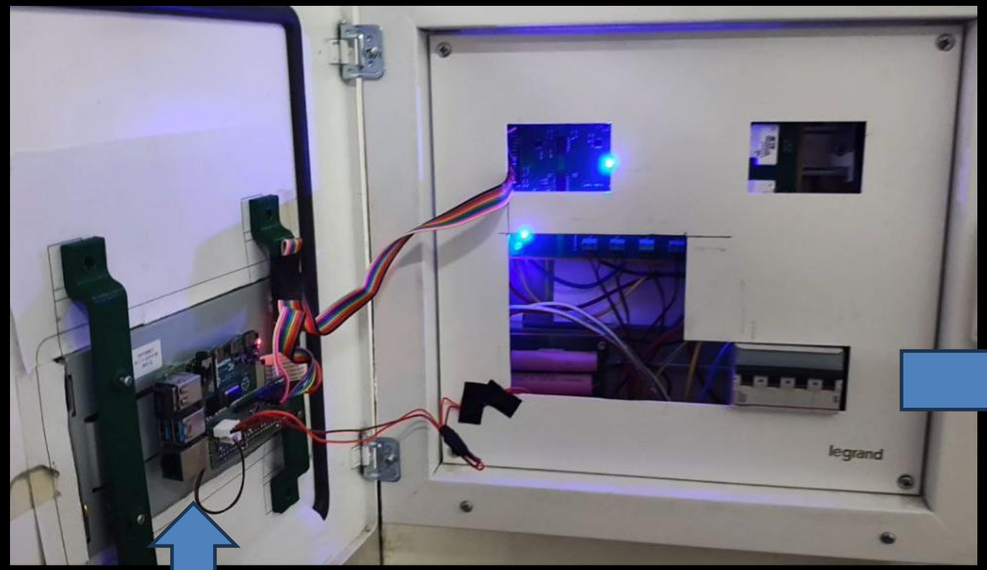
Current Voltage Phase R Phase Y Phase B

Current/Voltage

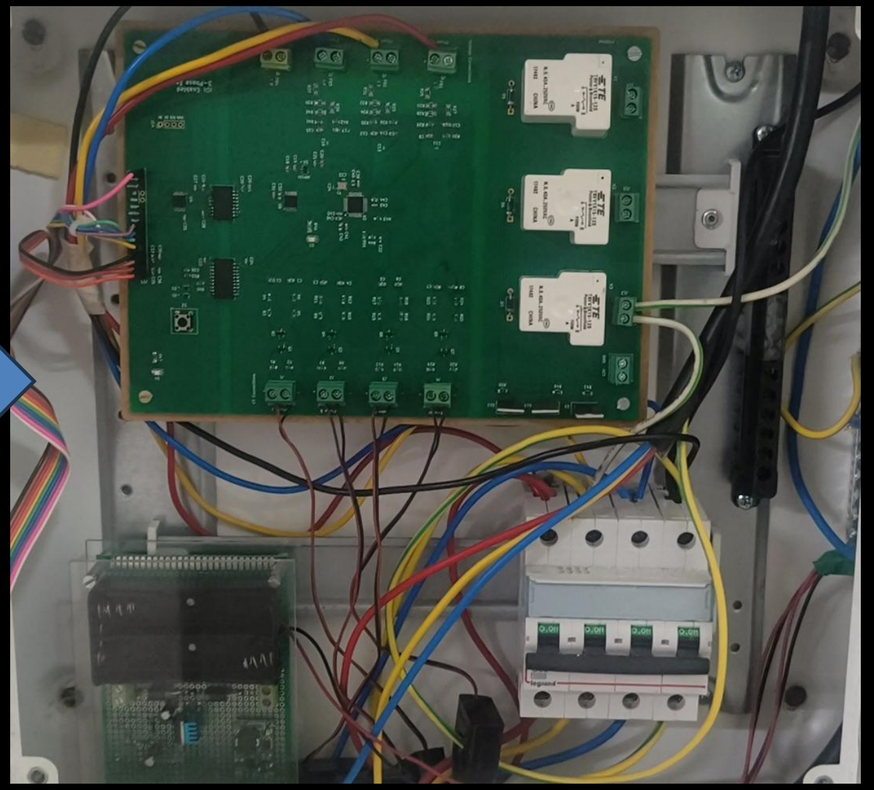
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00:00 03:00 06:00 09:00 12:00 15:00 18:00

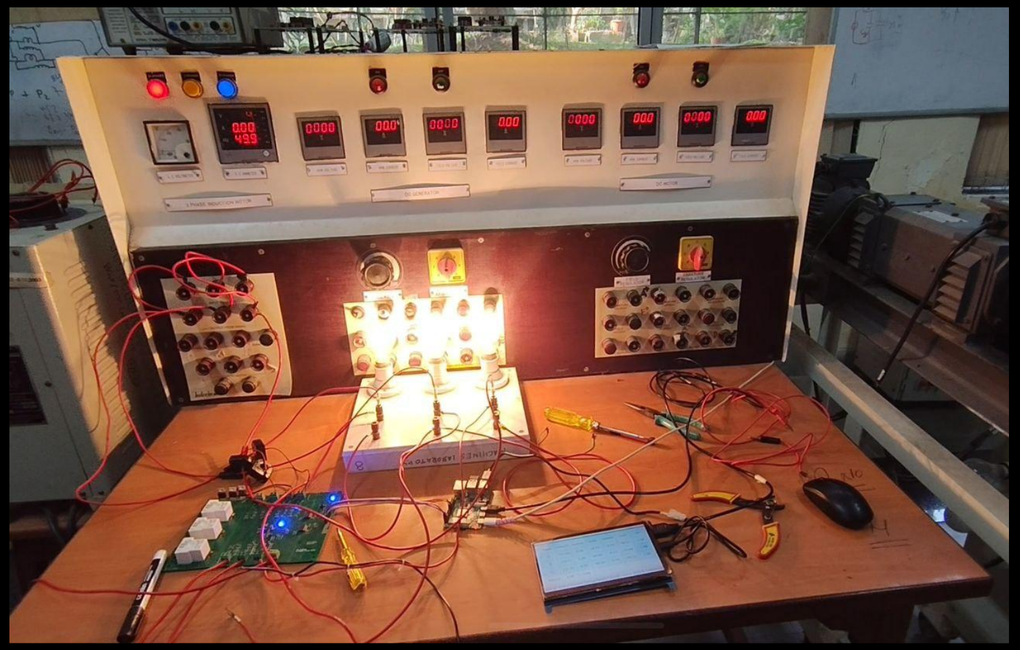
Energy | Power | Failure Log | Get QR | History | V - I | Home | Harmonics



Inside of this Casing



Display and RPI



Calibrated in Power Lab using 3-phase supply

Our Android application - VOLTIX

